

The Link Layer II & Network Synthesis

MPLS, Data Centers, and A Day in the Life

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Agenda: Wrapping Up the Wired World

- **Part 1: The Limits of Standard Routing**
- **Part 2: Multiprotocol Label Switching (MPLS)**
- **Part 3: Data Center Networking**
- **Part 4: Synthesis - A Day in the Life**

Recap: The Limits of Traditional IP Routing

Before we look at advanced networking, we must understand the limitations of the traditional Internet Protocol (IP) routing we studied in Chapter 5.

- **Destination-Only Forwarding:** A traditional IP router forwards a packet based strictly on the destination IP address. It does not care what kind of data is inside the packet.
- **Single Path Restriction:** Routing protocols like Open Shortest Path First (OSPF) calculate a single, least-cost path to a destination. All traffic is forced down that exact same path.
- **The Bottleneck:** If that single path gets congested, standard IP routing has no easy way to split the traffic and send half of it down a different, slightly longer path.

Introduction to MPLS (Section 6.5)

To solve the limitations of standard IP routing, internet backbone providers use a technology called **MPLS**.

- **Acronym Definition:** MPLS stands for **Multiprotocol Label Switching**.
- **The Core Concept:** Instead of forcing every router to perform complex lookups on long IP addresses, MPLS attaches a small, fixed-length "Label" to the packet.
- **Location in the Stack:** The MPLS label is inserted right between the Layer 2 (Ethernet) header and the Layer 3 (IP) header. Because of this, it is sometimes informally called a "Layer 2.5" protocol.

How MPLS Routers Work

An MPLS-capable router behaves very differently from a standard IP router.

- **Ignoring the IP:** When a packet arrives, the MPLS router completely ignores the destination IP address.
- **Reading the Label:** It simply reads the small MPLS label on the front of the packet.
- **The Swap and Forward:** It looks up that specific label in a very fast hardware table, swaps the old label for a new label, and immediately forwards the packet out the correct physical port.
- **The Result:** This makes forwarding significantly faster and requires less processing power than searching through massive IP routing tables.

The True Power of MPLS: Traffic Engineering

Speed is great, but the primary reason modern Internet Service Providers (ISPs) use MPLS is for **Traffic Engineering**.

- **Defining Custom Paths:** Network operators can pre-calculate exact, custom paths across their network and assign them specific MPLS labels, completely overriding OSPF.
- **Load Balancing:** If an ISP has two links to a city, they can use MPLS labels to force 50% of the web traffic down Link A, and 50% down Link B, preventing congestion.
- **Traffic Prioritization:** They can route standard email traffic over cheap, slow wires, and route premium voice-over-IP (VoIP) calls over fast, expensive fiber optic wires—even if both packets are going to the exact same destination IP!

Data Center Networking (Section 6.6)

When you search Google, watch Netflix, or use cloud software, your request does not go to a single computer. It goes to a massive **Data Center**.

- **The Massive Scale:** A modern mega-data center (like an Amazon Web Services facility) can house over 100,000 individual servers.
- **The Challenge:** Connecting 100,000 computers together so they can share data instantly requires highly specialized network architectures that look very different from a standard office network.
- **The Physical Layout:** Servers are not scattered randomly; they are stacked vertically in standardized metal cabinets called **Racks**. A single rack usually holds 20 to 40 blade servers.

The Top-of-Rack (TOR) Switch

How do the 40 servers in a single rack talk to the rest of the building?

- **Acronym Definition:** TOR stands for **Top-of-Rack**.
- **The Connection:** At the very top of each server rack sits a high-speed Link-Layer switch.
- **The Hierarchy:** Every single server in that specific rack plugs a short Ethernet cable directly into the TOR switch.
- **The Uplink:** The TOR switch then connects the entire rack of 40 servers to the broader data center network using highly redundant fiber optic cables.

Managing the Chaos: Load Balancers

How does a data center handle millions of simultaneous user requests hitting "www.google.com" at the exact same time?

- **The Public Face:** The outside world only sees one or two public IP addresses for the entire massive data center.
- **The Load Balancer:** When your HTTP request arrives at the data center, it first hits a specialized networking device called a **Load Balancer**.
- **Traffic Distribution:** The Load Balancer acts like a traffic cop. It looks at the thousands of internal servers, finds one that is currently not busy, and forwards your request to that specific machine's private internal IP address. This ensures no single server ever crashes from too much traffic.

Synthesis: A Day in the Life (Section 6.7)

To wrap up our study of the wired internet, let's trace a single, everyday action across the entire network protocol stack.

- **The Scenario:** You sit down at a coffee shop, open your laptop, and type `www.google.com` into your web browser.
- **What exactly happens?** We are going to trace the exact sequence of protocols required to make this work.
- **The Starting State:** Your laptop is turned on and connected to the local Wi-Fi. However, it currently has no IP address, no knowledge of the local router, and no knowledge of the outside Internet.

Step 1: Getting an Identity (DHCP)

Your laptop cannot send or receive any data on the Internet without an IP address.

- **Acronym Definition:** DHCP stands for **Dynamic Host Configuration Protocol**.
- **The Request:** Your laptop creates a DHCP Discover message. It wraps this message in a UDP segment, an IP datagram, and an Ethernet frame.
- **The Broadcast:** Because your laptop does not know where the DHCP server is, it sends the Ethernet frame to the broadcast MAC address (FF:FF:FF:FF:FF:FF) so every device in the coffee shop hears it.

Step 1 Continued: The DHCP Reply

The coffee shop's router hears your broadcast and responds.

- **The Offer:** The router's DHCP server sends back a DHCP ACK (Acknowledgment) message.
- **The Configuration:** This single message provides your laptop with four critical pieces of information:
 - 1 Your computer's new **IP Address**.
 - 2 The **Subnet Mask** (defining the local network size).
 - 3 The IP address of the **Default Gateway Router** (your exit to the Internet).
 - 4 The IP address of the **DNS Server** (the phonebook of the Internet).

Step 2: Finding the Server (DNS)

Your browser wants to connect to `www.google.com`, but computers only communicate using IP addresses.

- **Acronym Definition:** DNS stands for **Domain Name System**.
- **The Query:** Your laptop creates a DNS query asking, "What is the IP address for `google.com`?"
- **Addressing the Query:** It addresses this packet to the DNS Server IP address it received from DHCP just a moment ago.
- **The Roadblock:** Your laptop knows the router's IP address to send the packet out, but to put the frame on the physical Wi-Fi network, it needs the router's physical MAC address. It is currently stuck.

Step 3: Finding the Physical Path (ARP)

To solve this roadblock, we must use the Address Resolution Protocol.

- **Acronym Definition:** ARP stands for **Address Resolution Protocol**.
- **The ARP Request:** Your laptop pauses the DNS query. It broadcasts an ARP Request to the coffee shop: "Who has the IP address of the Default Gateway Router? Tell me your MAC address!"
- **The ARP Reply:** The router hears this and replies with its specific MAC address.
- **Moving Forward:** Your laptop finally writes the router's MAC address into the Ethernet frame and physically transmits the DNS query over the airwaves to the router.

Step 4: Routing Across the Globe (OSPF & BGP)

The router has your DNS query. How does it reach the DNS server miles away?

- **Leaving the LAN:** The router strips off your local Ethernet frame. It looks at the destination IP address (the DNS server) and consults its forwarding table.
- **Intra-AS Routing (OSPF):** The router uses the **Open Shortest Path First** protocol to navigate the packet efficiently through the local Internet Service Provider's network.
- **Inter-AS Routing (BGP):** To cross the global backbone between different competing ISPs, the packet is guided by the **Border Gateway Protocol**.
- **The DNS Reply:** The DNS server receives the query and routes a reply back to you: "Google's IP address is 142.250.190.46".

Step 5: Establishing the Connection (TCP)

Your laptop finally has Google's IP address! Now it can ask for the webpage.

- **Acronym Definition:** TCP stands for **Transmission Control Protocol**.
- **The Requirement:** Before sending sensitive HTTP data, your browser must establish a reliable, ordered, and error-checked connection with Google's server.
- **The 3-Way Handshake:**
 - 1 Your laptop sends a **SYN** (Synchronize) packet to Google.
 - 2 Google replies with a **SYN-ACK** (Synchronize-Acknowledge) packet.
 - 3 Your laptop sends a final **ACK** (Acknowledge) packet back.
- The secure connection is now officially open.

Step 6: Requesting the Data (HTTP)

The pipeline is built. It is time to get the website.

- **Acronym Definition:** HTTP stands for **Hypertext Transfer Protocol**.
- **The Request:** Your browser generates an **HTTP GET** request for the webpage's HTML file. It sends this request through the established TCP connection.
- **Data Center Processing:** The request arrives at Google's Data Center, hits the Load Balancer, and is forwarded to a specific blade server in a rack.
- **The Delivery:** The server packages the HTML and sends it back across the internet to your laptop.
- **Success:** The Google homepage renders on your screen. This entire 6-step process happened in roughly 200 milliseconds.

Questions?

We have successfully traced a packet from the Application Layer down to the physical wire and across the globe.